

1 · Gauss (+ superposition)

- $\oint \vec{E} \cdot d\vec{a} = Q_{enc}/\epsilon_0$. Pick a sphere, a coaxial cylinder or a pillbox so E is constant and $\parallel d\vec{a}$.
- Uniform ball: $\vec{E} = \frac{\rho}{3\epsilon_0}\vec{r}$ inside, $\frac{kQ}{r^2}\hat{r}$ outside. $V_{in} = \frac{kQ}{2R^3}(3R^2 - r^2)$, so $V(0) = \frac{3}{2}V(R)$.
- Infinite cylinder: $\vec{E} = \frac{\rho}{2\epsilon_0}s\hat{s}$ inside, $\frac{\lambda}{2\pi\epsilon_0}\hat{s}$ outside, with $\lambda = \rho\pi a^2$. Line: $V = -\frac{\lambda}{2\pi\epsilon_0}\ln\frac{s}{s_0}$. **Only** ΔV (it diverges at ∞).
- Shell: $E = 0$ inside. Slab/sheet: $\sigma/2\epsilon_0$. Conductor surface: σ/ϵ_0 .

- **Hole** = $+\rho + (-\rho)$. Use each piece's field from **its own** center/axis, then add the vectors.
- Spherical cavity in a ball, offset $\vec{d}$: $\vec{E} = \frac{\rho}{3\epsilon_0}\vec{d}$ (uniform). In a cylinder it is **not** uniform: $\frac{\rho}{\epsilon_0}(\frac{x}{6}, \frac{y}{6}, -\frac{z}{3})$.
- Ring (axis): $E_z = \frac{kqz}{(z^2 + R^2)^{3/2}}$. Disk: $E_z = \frac{\sigma}{2\epsilon_0}(1 - \frac{z}{\sqrt{z^2 + R^2}})$, $V = \frac{\sigma}{2\epsilon_0}(\sqrt{z^2 + R^2} - z)$.

2 · Given E (or V), find ρ : check the origin

- $\rho = \frac{\epsilon_0}{r^2} \frac{d}{dr}(r^2 E_r)$ for $r \neq 0$. Then $Q_{enc}(r) = 4\pi\epsilon_0 r^2 E_r$.
- If $\lim_{r \rightarrow 0} 4\pi\epsilon_0 r^2 E_r = q \neq 0$: point charge, $\rho \propto q\delta^3(\vec{r})$. ($E \sim 1/r^2$ near 0 $\Rightarrow$ there is a δ .)
- $\nabla \cdot \frac{\vec{r}}{r^3} = 4\pi\delta^3(\vec{r})$, $\nabla^2 \frac{1}{r} = -4\pi\delta^3(\vec{r})$, Poisson $\nabla^2 V = -\rho/\epsilon_0$.
- $Q_{tot} = \lim_{r \rightarrow \infty} 4\pi\epsilon_0 r^2 E_r$. Check: $Q_{tot} = q + \int \rho d\tau$ (+ any σ from jumps in E_r).
- $V = \frac{Ae^{-\lambda r}}{r} \Rightarrow \rho = \epsilon_0 A[4\pi\delta^3(\vec{r}) - \lambda^2 \frac{e^{-\lambda r}}{r}]$.
- $E \propto r^{-4}$ all the way to $r = 0 \Rightarrow$ infinite charge (not a δ). A sensible version has $1/r^4$ only for $r > R$, or a mix like $\frac{A}{r^2} + \dots$.
- $\int f\delta(x-a)dx = f(a)$, $\delta(kx) = \delta(x)/|k|$. Shell: $\rho = \sigma\delta(r-R)$.

3 · Boundary conditions (number them!)

- Across σ : $E_{above}^\perp - E_{below}^\perp = \sigma/\epsilon_0$. $E^\parallel$ and V continuous. $\frac{\partial V_{out}}{\partial n} - \frac{\partial V_{in}}{\partial n} = -\frac{\sigma}{\epsilon_0}$.
- Conductor: $E = 0$ and $\rho = 0$ inside, V const, $\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$, $\vec{E}_{out} = \frac{\sigma}{\epsilon_0}\hat{n}$, pressure $\frac{\sigma^2}{2\epsilon_0}$.
- Checklist: (1) each surface: V given, or $V = \text{const}$ with Q given. (2) Finite at $r = 0$ / $s = 0$. (3) $V \rightarrow 0$ at ∞ (or $\rightarrow -E_0 r \cos\theta$). (4) Continuity + jump at each shell. (5) Symmetry.
- "Grounded" = $V = 0$. "Isolated, neutral" = V unknown const + $\oint \sigma = 0$. "Held at V_0 " = battery: Q free.
- Uniqueness: V (or Q on each conductor) on every boundary + $\rho \Rightarrow$ unique. Any guess that fits **is** the answer.
- Laplace: no max/min inside, $V(\text{center}) =$ average over a sphere. Earnshaw: no stable equilibrium.

4 · Images

- Plane, q at height d : $-q$ at $-d$. $\sigma = \frac{-qd}{2\pi(s^2 + d^2)^{3/2}}$, total $-q$; $F = \frac{kq^2}{(2d)^2}$ (attract). Half the charge lies within $s = \sqrt{3}d$.
- **Energy = half** of the real + image pair: $W = -\frac{kq^2}{4d}$. Work to pull q off: $+\frac{kq^2}{4d}$.
- 90° corner: 3 images ($-q, -q, +q$). Wedge π/n : $2n - 1$. Two parallel planes: ∞ row. Line λ over a plane: $-\lambda$.
- **Grounded sphere** (q at $a > R$): $q' = -\frac{R}{a}q$ at $b = \frac{R^2}{a}$. $F = \frac{kq^2 Ra}{(a^2 - R^2)^2}$ (attract). $\sigma(\theta) = \frac{-q(a^2 - R^2)}{4\pi R(R^2 + a^2 - 2aR\cos\theta)^{3/2}}$, total $-qR/a$.
- Twists = add a charge at the **center**: held at V_0 : $+4\pi\epsilon_0 R V_0$. Neutral: $+\frac{R}{a}q$. Total charge Q : $Q + \frac{R}{a}q$.
- q **inside** a shell at distance d : image $-\frac{R}{d}q$ at $\frac{R^2}{d}$ (outside). Shell at V_0 : just add V_0 inside (no force change).
- Image valid **only** in the region of interest (outside the metal). Images never sit in that region.

5 · Energy, capacitors

- $W = \frac{1}{2} \sum q_i V_i$ (V from the **others**) = $\frac{1}{2} \int \rho V d\tau = \frac{\epsilon_0}{2} \int E^2 d\tau$. Not superposable (cross terms).
- Uniform ball: $\frac{3}{5} \frac{kQ^2}{R}$. Shell: $\frac{1}{2} \frac{kQ^2}{R}$. Pair: $\frac{kq_1 q_2}{r}$.
- $C = Q/\Delta V$, $W = \frac{1}{2} C V^2 = \frac{Q^2}{2C}$. Plates $\frac{\epsilon_0 A}{d}$; spheres $\frac{4\pi\epsilon_0 ab}{b-a}$; coax $\frac{2\pi\epsilon_0 L}{\ln(b/a)}$.

6 · Separation: Cartesian

- Direction with **two zero** BCs $\Rightarrow \sin \frac{n\pi y}{a}$ (oscillates); the other $\Rightarrow e^{\pm kx}$, $\sinh$, $\cosh$, with $k = \frac{n\pi}{a}$.
- Open to ∞ : e^{-kx} . Zero at $x = b$: $\sinh k(b-x)$. Zero at 0: $\sinh kx$. Symmetric: $\cosh k(x - \frac{b}{2})$.
- Fourier: $\int_0^a \sin \frac{n\pi y}{a} \sin \frac{m\pi y}{a} dy = \frac{a}{2} \delta_{nm}$. $C_n = \frac{2}{a} \int_0^a V_0(y) \sin \frac{n\pi y}{a} dy$. Const V_0 : $\frac{4V_0}{n\pi}$, odd n .
- Live face at $x = b$: $V = \sum C_n \sin \frac{n\pi y}{a} \frac{\sinh(n\pi x/a)}{\sinh(n\pi b/a)}$. Several live faces: superpose one per face.
- Slot: $V = \frac{4V_0}{\pi} \sum_{\text{odd } n} \frac{1}{n} e^{-n\pi x/a} \sin \frac{n\pi y}{a} = \frac{2V_0}{\pi} \tan^{-1} \frac{\sinh(\pi y/a)}{\sinh(\pi x/a)}$.
- 3-D: $\sin \sin \times \sinh(\pi \sqrt{n^2/a^2 + m^2/b^2} z)$; const: $C_{nm} = \frac{16V_0}{\pi^2 nm}$ (odd, odd).
- Series $\rightarrow V_0/2$ at a discontinuity; Gibbs overshoot only in the partial sum.

7 · Separation: polar (2-D, no z)

- $\nabla^2 = \frac{1}{s} \partial_s(s \partial_s) + \frac{1}{s^2} \partial_\phi^2$. Try $R = s^\lambda \Rightarrow \lambda = \pm k$.
- $V = a_0 + b_0 \ln s + \sum_k [s^k (a_k \cos k\phi + b_k \sin k\phi) + s^{-k} (c_k \cos k\phi + d_k \sin k\phi)]$.
- Full circle $\Rightarrow k$ integer, no ϕ term. Origin in region $\Rightarrow$ drop s^{-k} and $\ln s$. $\infty \Rightarrow$ drop s^k ($\ln s \leftrightarrow$ net λ).
- **Wedge** walls $\theta = 0, \alpha$ at 0: $\sin \frac{n\pi\theta}{\alpha}$, $k = \frac{n\pi}{\alpha}$. Zero arc at $r = a$: $(\frac{r}{a})^k - (\frac{a}{r})^k$. Live arc $r = b$ at V_0 : $C_n = \frac{2V_0(1-(-1)^n)}{n\pi[(b/a)^k - (a/b)^k]}$ (Spring '26).
- One wall at V_0 , the other 0, no arcs: $V = V_0\theta/\alpha$. Tip field $\propto r^{\pi/\alpha-1}$ (blows up if $\alpha > \pi$).
- Pipe in $E_0\hat{x}$: $V = -E_0(s - \frac{R^2}{s}) \cos\phi$, $\sigma = 2\epsilon_0 E_0 \cos\phi$.

8 · Separation: spherical / Legendre

- $V = \sum (A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}) P_\ell(\cos\theta)$. Inside: A only. Outside: B only ($+ -E_0 r \cos\theta$). $r^{-(\ell+1)}$, **not** $r^{-\ell}$.
- $P_0 = 1$, $P_1 = x$, $P_2 = \frac{3x^2-1}{2}$, $P_3 = \frac{5x^3-3x}{2}$, $P_4 = \frac{35x^4-30x^2+3}{8}$. $P_\ell(1) = 1$, $P_\ell(-1) = (-1)^\ell$, $P_2(0) = -\frac{1}{2}$, $P_4(0) = \frac{3}{8}$.
- $x^2 = \frac{1}{3} + \frac{2}{3}P_2$, $x^3 = \frac{3}{5}P_1 + \frac{2}{5}P_3$, $x^4 = \frac{1}{5} + \frac{4}{7}P_2 + \frac{8}{35}P_4$, $\cos 2\theta = -\frac{1}{3} + \frac{4}{3}P_2$, $\sin^2\theta = \frac{2}{3} - \frac{2}{3}P_2$.
- $\int_{-1}^1 P_\ell P_m dx = \frac{2\delta_{\ell m}}{2\ell+1} \Rightarrow c_\ell = \frac{2\ell+1}{2} \int_{-1}^1 f(x) P_\ell dx$ ($dx = \sin\theta d\theta$).
- Shell at $V_0(\theta) = \sum c_\ell P_\ell$: in $c_\ell(\frac{r}{R})^\ell$, out $c_\ell(\frac{R}{r})^{\ell+1}$. $\sigma = \frac{\epsilon_0}{R} \sum (2\ell+1) c_\ell P_\ell$, $Q = 4\pi\epsilon_0 R c_0$.
- Given $\sigma = \sum \sigma_\ell P_\ell$: $A_\ell = \frac{\sigma_\ell}{(2\ell+1)\epsilon_0 R^{\ell-1}}$, $B_\ell = A_\ell R^{2\ell+1}$.
- Metal sphere in $E_0\hat{z}$: $V = -E_0(r - \frac{R^2}{r^2}) \cos\theta$, $\sigma = 3\epsilon_0 E_0 \cos\theta$ ($+\frac{R V_0}{r}$ if at V_0).
- **Axis trick**: know $V(z)$ on the axis, expand in z^ℓ or $z^{-(\ell+1)}$, replace by $r^\ell P_\ell$ or $r^{-(\ell+1)} P_\ell$.
- $\frac{1}{|\vec{r}-\vec{r}'|} = \sum \frac{r_\ell^r}{r_\ell^r} P_\ell(\cos\gamma)$.

9 · Multipoles

- $V = k[\frac{Q}{r} + \frac{\vec{p} \cdot \vec{r}}{r^3} + \dots]$, $V_\ell = \frac{k}{r^{\ell+1}} \int r'^\ell P_\ell(\cos\theta') \rho d\tau'$. $\vec{p} = \int \rho \vec{r}' d\tau' = \sum q_i \vec{r}_i$.
- $\vec{E}_{dip} = \frac{k\vec{p}}{r^3} (2\cos\theta \hat{r} + \sin\theta \hat{\theta}) = \frac{k}{r^3} [3(\vec{p} \cdot \hat{r})\hat{r} - \vec{p}]$.
- $\rho(\phi)$ -free $\Rightarrow$ only $p_z = 2\pi \iint \rho r'^3 \sin\theta' \cos\theta' dr' d\theta'$ (Spring '26: $\pm\rho_0$ halves $\Rightarrow \frac{\pi R^2 \rho_0}{2}$).
- Odd in $z \Rightarrow$ only odd ℓ . Even $\Rightarrow$ even ℓ . $Q \neq 0 \Rightarrow \vec{p}$ depends on origin: $\vec{p}' = \vec{p} - Q\vec{a}$.
- $\vec{N} = \vec{p} \times \vec{E}$, $U = -\vec{p} \cdot \vec{E}$, $\vec{F} = (\vec{p} \cdot \nabla)\vec{E}$. V falls as r^{-1}, r^{-2}, r^{-3} ; E one power faster.

10 · Classic results

- Shell $\sigma = k \cos\theta$: $V_{in} = \frac{k}{3\epsilon_0} r \cos\theta$ (uniform $\vec{E} = -\frac{k}{3\epsilon_0}\hat{z}$); outside a pure dipole $p = \frac{4}{3}\pi R^3 k$.
- Nested metal shells: inner surface carries $-q_{enc}$, outer surface carries $Q_{shell} + q_{enc}$. A cavity's field depends only on the charge inside it.
- Edge at V_0 : $C_n = \frac{2V_0(-1)^{n+1}}{n\pi}$. Const: $\frac{4V_0}{n\pi}$ (odd). $\int_0^a \sin^2 = \frac{a}{2}$.
- Line λ at height d over a grounded plane: $V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{s_-}{s_+}$ ($s_\pm =$ distance to $\pm\lambda$).
- Dipole: $V = \frac{k p \cos\theta}{r^2}$. Ball $\rho_0 \cos\theta$ and shell $k \cos\theta$: exact dipole outside.

11 · Which method? Traps

- High symmetry $\Rightarrow$ Gauss. Charge + flat/spherical metal $\Rightarrow$ images. Box/slot $\Rightarrow$ Cartesian. Sphere with $V(\theta)$ or $\sigma(\theta) \Rightarrow$ Legendre. Wedge/pipe $\Rightarrow$ polar.
- Setup gets most of the points: region, numbered BCs, general solution **first**.
- $d\tau = r^2 \sin\theta dr d\theta d\phi$; σ/ϵ_0 (**metal**) vs $\sigma/2\epsilon_0$ (**sheet**); images **half energy**; check units and limits.
- $\int \frac{dx}{(x^2+a^2)^{3/2}} = \frac{x}{a^2 \sqrt{x^2+a^2}}$, $\int \frac{x dx}{(x^2+a^2)^{3/2}} = \frac{-1}{\sqrt{x^2+a^2}}$, $\int \frac{dx}{\sqrt{x^2+a^2}} = \ln(x + \sqrt{x^2+a^2})$, $\int_0^\pi \sin^3 = \frac{4}{3}$, $\int_0^\pi \sin\theta \cos^2\theta = \frac{2}{3}$.